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Research paper

Quantitative assessment of static voltage stability for power system with high-penetration wind power based on energy function

Jian Zhang [a], Ruicong Ma [a], Yongji Cao [b,*], Shihao Zou [a], Changgang Li [a]

[a] Key Laboratory of Power System Intelligent Dispatch and Control of the Ministry of Education, Shandong University, Jinan 250061, China
[b] Academy of Intelligent Innovation, Shandong University, Jinan 250101, China

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ABSTRACT

As the scale of power systems expands, the penetration of renewable energy sources and power electronic devices are accelerating. With the growing complexity of power system, there is an urgent need for in-depth research on the impact of reactive power on static voltage stability. This paper proposes a quantitative assessment approach of static voltage stability for the power system with high-penetration wind power based on the energy function. A quantitative assessment framework for voltage stability indices based on power flow analysis has been established. The classic static voltage stability index L based on reactive power has been comprehensively analyzed. Additionally, combining the energy function and power flow analysis, a new quantitative index for voltage stability based on bus reactive power injection has been proposed. Finally, the effectiveness of this index has been validated by case studies. With the increasing penetration of wind power, the trend of the change of the assessment index of voltage margin is analyzed.

1. Introduction

The management of energy flow and power allocation in power systems is essential to all issues within the field of power systems and constitutes a crucial research focus in the planning and operation of power systems (Cao et al., 2024, 2023; Zhang et al., 2023). With the continuous expansion of power grid scale and the increasing complexity of grid structures, along with significant interconnection characteristics among grid regions, there is a need for flexible quantitative metrics to conduct zoning studies on the voltage stability of power systems (Seyedi et al., 2022; Cao et al., 2022; Taheri et al., 2023). In order to meet the demands of an expanding power grid, it is of the utmost importance to focus on the distribution of reactive power within the grid. It is equally vital to further investigate the fundamental physical mechanisms, distribution patterns, and their consequential impacts on grid states (Veunnasack et al., 2023; Shahriyari et al., 2023; Zhang et al., 2024).

The concept of reactive power, which has been extensively researched and refined, is currently defined as the maximum energy exchange between internal and external elements of the grid (Ghosh and Moursi, 2022). It has been noted that in certain circuits, such as resonant circuits, the maximum energy exchange between inductors and capacitors and external sources does not necessarily correspond to reactive power (Pan et al., 2020). Therefore, further investigation into the mechanisms and physical essence behind the generation of reactive power is warranted. Additionally, a method for zoning reactive power control in power systems based on power flow tracking and clustering algorithms has been proposed (Deng et al., 2022a). Expanding on the aforementioned research, the utilization of regional reactive power reserves to tackle the issue of reactive power imbalance within specific regions is examined, and the model for zoning reactive power voltage optimization is refined (Li et al., 2021a; Pei et al., 2020; Vineeth et al., 2024).

Concurrently, in pursuit of strategic objectives such as reaching peak carbon emissions and achieving carbon neutrality, the extensive integration of green renewable energy into the grid has emerged as a pivotal focus for the advancement of the Chinese power grid, particularly with wind energy serving as a rapidly expanding exemplar (Park et al., 2021; Lv et al., 2023). Presently, wind energy technology development and commercial potential in China are progressively maturing, leading to a gradual decrease in the production cost of wind power. In certain regions, the cost of wind power generation has already dipped below that of conventional synchronous generators (Matavalam et al., 2022; Mardanimajd et al., 2024).

Simultaneously, in pursuit of strategic objectives such as reaching

* Corresponding author.
E-mail address: yongji@sdu.edu.cn (Y. Cao).

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peak carbon emissions and achieving carbon neutrality, the extensive integration of green renewable energy into the grid has emerged as a pivotal focus for the advancement of the Chinese power grid, particularly with wind energy serving as a rapidly expanding exemplar (Park et al., 2021; Lv et al., 2023). Presently, wind energy technology development and commercial potential in China are progressively maturing, leading to a gradual decrease in the production cost of wind power. In certain regions, the cost of wind power generation has already dipped below that of conventional synchronous generators (Matavalam et al., 2022; Mardanimajd et al., 2024).

In order to prevent power accidents, grid dispatchers are primarily concerned with the operational and voltage stability of the current system. Specifically, their focus lies in determining the distance of the voltage stability margin from the critical point (Guo, 2020). Therefore, there is a necessity for a suitable metric to evaluate the voltage stability margin, which will provide operators with a reference point for understanding the operational status of the system and taking appropriate actions (Alzubaidi et al., 2024). The voltage stability assessment indices quantitatively measure the distance between the current state of the system and its critical state, thereby indicating the level of voltage stability. Different methods for assessing voltage stability yield various indices, which can be used to determine the level of voltage stability under different operational conditions. Up to now, scholars from both domestic and international arenas have put forward a range of voltage stability margin indices from diverse perspectives, encompassing sensitivity indices, eigenvalue and eigenvector indices, load margin indices, as well as L indices (Deng et al., 2022b; Chen et al., 2022; Bharati and Ajjarapu, 2020).

However, while wind power generation offers significant advantages, it also exacerbates concerns regarding the security and stability of the power grid (Liu et al., 2024; Yun and Cui, 2020; Zhang et al., 2022a). The intermittency of wind energy presents challenges to the security and stability of the grid due to its inherent randomness, volatility, and anti-peak characteristics. Extreme weather conditions can have a substantial impact on wind turbine output, and large-scale long-distance transmission of wind power may pose certain risks to the power system. The fluctuating nature of turbine output may negatively affect the balance, stability, and security of power generation in specific areas of the grid (Van Cutsem et al., 2020; Me et al., 2023; Li et al., 2021b). These factors must be comprehensively taken into account. In the normal operation of the system, it is essential to establish an index for assessing the static voltage stability of the power system (Zuo et al., 2023; Kim et al., 2021; Jia et al., 2024). This offers a relatively intuitive indication of system status for planning and scheduling operations, identifies weak areas within the system, and helps prevent emergencies (Qi et al., 2019). Therefore, obtaining an accurate voltage stability margin index is crucial for evaluating the degree of stability within the system (Zhang et al., 2022b; Gurusinghe and Rajapakse, 2016).

Upon surveying the current methods for voltage stability, it is evident that there are several approaches. The continuous power flow method is used for analysis and the gradient vector index is utilized for quadratic curve fitting (Zhang et al., 2005). This approach enables swift computation of state variables and variable parameters during prediction. However, it still faces inherent limitations such as poor convergence at critical points and instances of reactive power constraint violations. On the other hand, the sensitivity is calculated through the linear relationship between voltage stability margin and parameters to analyze stability (Yan and Saha, 2012). This method primarily involves qualitative research, posing challenges for quantitative analysis. Furthermore, this approach does not account for situations where reactive power constraints are violated. The singular value analysis is combined with sensitivity analysis to develop an index for identifying weak buses (Wang and Girgis, 1996). However, due to the prevalence of nonlinear components in the system, singular value decomposition is employed post-system linearization, resulting in inevitable errors given the nonlinear nature of actual power systems. In (Li et al., 2018), a quantitative index is formulated based on the power flow multi-solution method for determining static voltage stability. Nevertheless, this approach suffers from high computational complexity, posing challenges for its implementation in online operation. The probabilistic eigen analysis method is incorporated into small disturbance analysis to determine the probabilistic critical load level and assess stability based on eigen distribution (Zhu and Liu, 2016). However, the small-disturbance analysis method utilizing linearized dynamic equations encounters challenges such as intricate differential equation analysis, high matrix dimensions, and complexities in eigenvalue analysis. Based on the real-time measurement data and deep transfer learning, short-term voltage stability assessment of power systems was analyzed (Li et al., 2023). Then, the voltage stability of the multi-send high-voltage direct current (HVDC) system has been quantified, and the impacts of HVDC transmission capacity has been analyzed (Wang et al., 2022). In addition, the deep learning based on transformer architecture has been used for the power system short-term voltage stability assessment (Li et al., 2024).

Therefore, by integrating the energy function and power flow analysis, a quantitative metric for voltage stability based on reactive power injection at bus bars has been introduced. The correlation between reactive power reserve and the voltage stability index is examined. Subsequently, a comprehensive assessment framework for voltage stability indices is established, with a thorough analysis of the classic static voltage stability index L based on reactive power. Through the combination of energy function and power flow analysis, a quantitative metric for static voltage stability has been proposed.

Compared to traditional methods, the proposed assessment approach has several advantages to tackle the variability and intermittency of wind energies, which can be listed as follows:

(1) The traditional L index relies on linearization analysis, which produces large errors when the system tends to a critical state. The proposed energy function-based method can better deal with the nonlinear characteristics of the system and reflect the real system state under wind fluctuation more accurately.
(2) The proposed assessment framework can carry out dynamic analysis based on real-time data. The scheme can quickly identify and respond to voltage stability changes by power flow analysis. Thus, it provides higher sensitivity and accuracy.
(3) The proposed framework focuses on not only reactive power injection and consumption, but also the system topology and the variations of load and wind power. Thus, the scheme can provide a more comprehensive assessment of voltage stability, especially for the key buses around wind power integration points.

The remainder of this manuscript is structured as follows: Section 2 examines the correlation between reactive power reserve and voltage stability. Additionally, based on the power flow analysis, Section 3 introduces a scheme for assessing voltage stability. In Section 4, an analysis of the L index and the proposed energy function-based voltage stability index is presented. A case study is conducted in Section 5. Finally, conclusions and discussions are provided in Section 6.

2. Relationship between reactive power reserve and voltage stability

The generation of reactive power in power systems arises from the necessity for electric motors to establish and sustain a rotating magnetic field, which facilitates rotor rotation and mechanical motion. The establishment of the rotor's magnetic field in electric motors is reliant upon acquiring reactive power from the power source. Consequently, modulation of the excitation current can effectively control reactive power, thereby adjusting the intensity of the magnetic field.

The voltage magnitude at a specified load bus serves as the most straightforward indicator for evaluating the occurrence of voltage collapse at that particular bus. In order to analyze the voltages, it is possible to compute the power flow as,

$$\begin{cases} S_i \cos\varphi = \dfrac{U_i}{Z}\left[E\cos\left(\theta - \left(\delta_E - \delta_i\right)\right) - U_i\cos\theta\right] \\ S_i \sin\varphi = \dfrac{U_i}{Z}\left[E\sin\left(\theta - \left(\delta_E - \delta_i\right)\right) - U_i\sin\theta\right] \end{cases} \tag{1}$$

where E is the equivalent potential amplitude of the load bus, δ_E is the equivalent phase angle of the load bus, Z is the equivalent impedance modulus of the load bus, θ is the equivalent impedance phase angle of the load bus, S_i is the apparent power of the load bus, and φ is the power factor angle of the load bus.

From the above equation, we can eliminate (δ_E -δ_i) and get the relationship equation between U_i and S_i as,

$$\left(U_i^2\right)^2 + \left[2ZS_i\cos(\theta-\varphi) - E^2\right]U_i^2 + Z^2S_i^2 = 0 \tag{2}$$

Due to the nonlinearity and multiple solutions of the current equation, when the system voltage is stable, there are two solutions for the current equation. The system is generally considered to be at the voltage stabilization limit point when there is a re-root, indicating the following relation,

$$\frac{E^4}{4} - Z^2S_i^2\sin^2\left(\theta-\varphi\right) - E^2ZS_i\cos\left(\theta-\varphi\right) = 0 \tag{3}$$

Thus, the critical expression for the bus voltage $U_{cr,i}$ at this point can be derived as,

$$U_{cr,i} = \frac{E}{\sqrt{2[1+\cos(\theta-\varphi)]}} \tag{4}$$

Based on this foundation, the normalized analytical voltage margin index is obtained as,

$$K_{U,i} = \frac{U_i - U_{cr,i}}{U_i} \times 100\% = \left(1 - \frac{E}{\sqrt{2[1+\cos(\theta-\varphi)]}^{*}U_i}\right) \times 100\% \tag{5}$$

When this load bus voltage is greater than the critical stabilization voltage, i.e., $0<K_{U,i}<100\%$, this bus voltage is stable. And the larger value of $K_{U,i}$ indicates the larger voltage stabilization margin. At the voltage stabilization limit point $K_{U,i}=0$, the bus voltage is critically stable.

The apparent power transmitted from the grid to the load bus is maximized when (4) is satisfied, which can be obtained by bringing (5) into (2), and expressed as,

$$S_{\max,i} = \frac{E^2}{2Z[1+\cos(\theta-\varphi)]} \tag{6}$$

Therefore, the normalized analytical power margin index is obtained as:

$$K_{S,i} = \frac{S_{\max,i} - S_i}{S_{\max,i}} \times 100\% = \left\{1 - \frac{2S_iZ[1+\cos(\theta-\varphi)]}{E^2}\right\} \times 100\% \tag{7}$$

When the current power transmitted by the system to this load bus is less than the maximum transmission power that the system can provide ($0<K_{S,i}<100$ %), the voltage at this bus becomes stabilized, with a larger value of $K_{S,i}$ corresponding to a greater voltage stabilization margin. When the current power transmitted by the system to the load bus equals the maximum transmission power that the system can provide ($K_{S,i}=0$), the voltage of the bus reaches its limit point of stabilization.

The relationship between voltage stability margin and sufficient reactive reserve in power systems is of utmost importance for the enhancement of grid voltage stability. The management of reactive reserve has the potential to effectively increase the voltage stability margin, although its impact is contingent upon the power system network topology and load growth scenarios. Specifically, the efficacy of reactive reserve management may manifest in a linear or quadratic form, depending on the specific characteristics and operational scenarios of the power system. Through an examination of multivariate regression models, a more comprehensive understanding of the correlation between reactive reserve and voltage stability margin can be attained, providing theoretical underpinning and guidance for grid stability. This research approach systematically analyzes the influence of reactive reserve on voltage stability margin under varying conditions, offering valuable insights for power system operation and management.

Currently, there is no definitive quantitative relationship between reactive power reserve and voltage stability margin in the existing research. A multilinear regression model can be used to establish a quantitative functional relationship, which is expressed as,

$$y_i = a_0 + a_jx_{ij} + \gamma_kx_{ik}x_{il} + \omega_jx_{ij}^2 + \varepsilon_i \tag{8}$$

where the vector y represents the voltage stability margin of the system. This model is linear in coefficients α, γ, and ω, and solved using the least squares method. When there are several observations, the coefficient vector β can be expressed as,

$$\beta = \left[a_0, \ldots, a_p, \gamma_1, \ldots, \gamma_{p(p-1)/2}, \ldots, \omega_1, \ldots, \omega_p\right]^T \tag{9}$$

$$y = \boldsymbol{X}\beta + \varepsilon \tag{10}$$

where the matrix $\boldsymbol{X}$ represents the matrix of reactive reserve for the system. Each element value in the matrix y represents a sample of the system's voltage stability margin.

The reactive reserve and voltage stability margin samples are subsequently collected at various points along the PV curve, enabling the establishment of a quantitative relationship between these two variables within this model. The coefficient vector β is then derived by minimizing the sum of squared residuals, which can be expressed as,

$$Min \parallel \varepsilon\parallel^2 = Min \parallel y - \boldsymbol{X}\beta\parallel^2 \tag{11}$$

The iterative process leads to the optimal linear unbiased estimate of the coefficient vector β:

$$\widehat{\beta} = \left(\boldsymbol{X}^T\boldsymbol{X}\right)^{-1}\left(\boldsymbol{X}^Ty\right) \tag{12}$$

The quantitative calculation between reactive power reserve and voltage stability margin is subsequently determined by employing multiple linear regression equations based on the aforementioned solution results, and expressed as,

$$\widehat{y} = \boldsymbol{X}\widehat{\beta} \tag{13}$$

The relationship between voltage stability margin and reactive reserve can be expressed as,

$$VSM = k\sum_{i=1}^{n}\partial_iQ_{ri} + b \tag{14}$$

where k represents the slope of the curve depicting the relationship between voltage stability margin and reactive reserve, $\sum_{i=1}^{n}\partial_iQ_{ri}$ denotes the reactive power reserve, b signifies the confidence interval for voltage stability margin, n indicates the number of key wind turbines. The overall voltage stability margin of the power system demonstrates a linear correlation with the cumulative reactive reserve of the system.

3. Voltage Stability Assessment Based on Power Flow Analysis

Voltage stability assessment methods can be broadly categorized into static and non-static approaches. Static methods rely on network flow equations, while non-static methods are based on network state equations. Compared to non-static approaches, static methods require less computational effort and primarily reflect the current level of system stability. They offer various indices of system stability and preventative

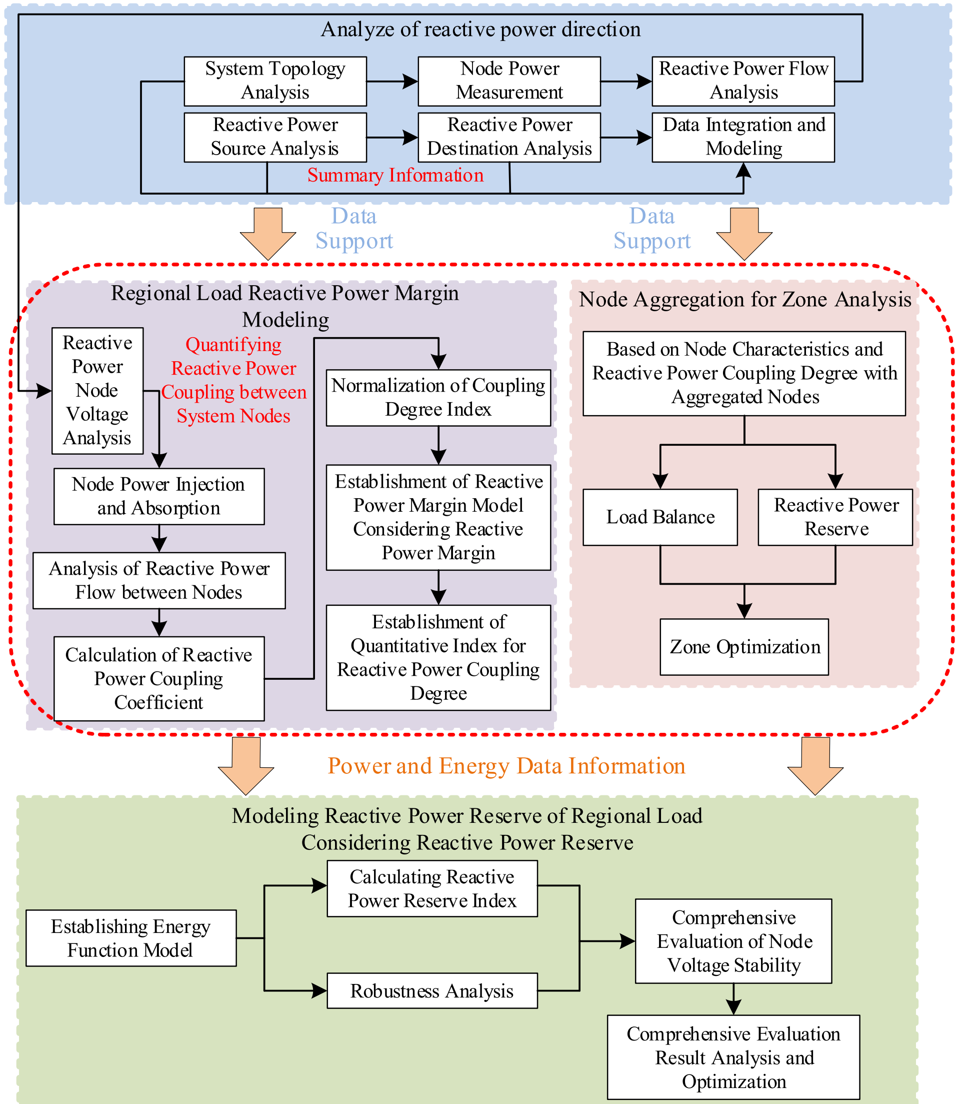

Fig. 1. Schematic diagram of the proposed voltage stability assessment approach.

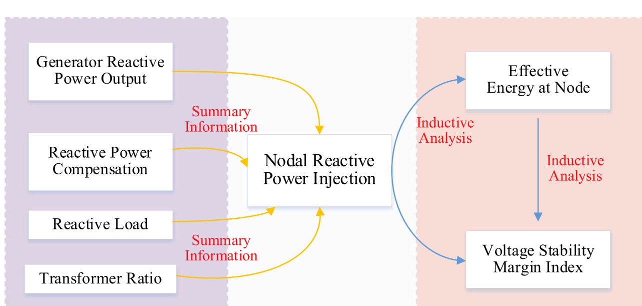

Fig. 2. Schematic diagram of energy function model.

measures, such as distance to the stability boundary and sensitivity information regarding state and control variables. This analytical approach includes sensitivity analysis, continuation power flow method, eigenvalue analysis method, singular value analysis method, multi-solution power flow method, nonlinear programming method etc.

This paper proposes the subsequent design process for quantifying and assessing voltage stability based on power flow analysis, of which the procedures can be shown in Fig. 1. Firstly, real-time data of each bus

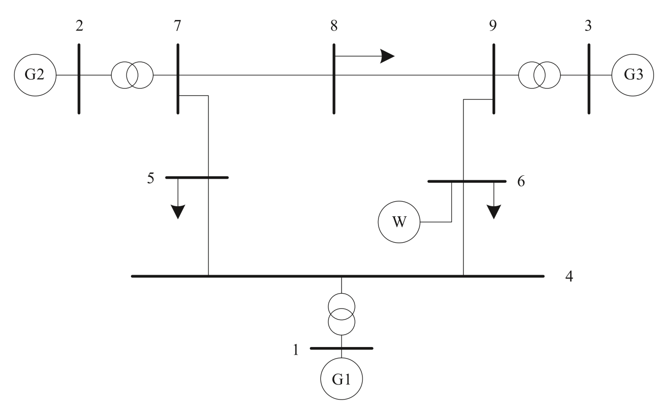

Fig. 3. Schematic diagram of test system.

in the system, including voltage, reactive power, and active power information, are collected. Subsequently, a topological analysis is conducted to ascertain the interconnection and power transmission pathways between buses.

Subsequently, a reactive power flow analysis is conducted to

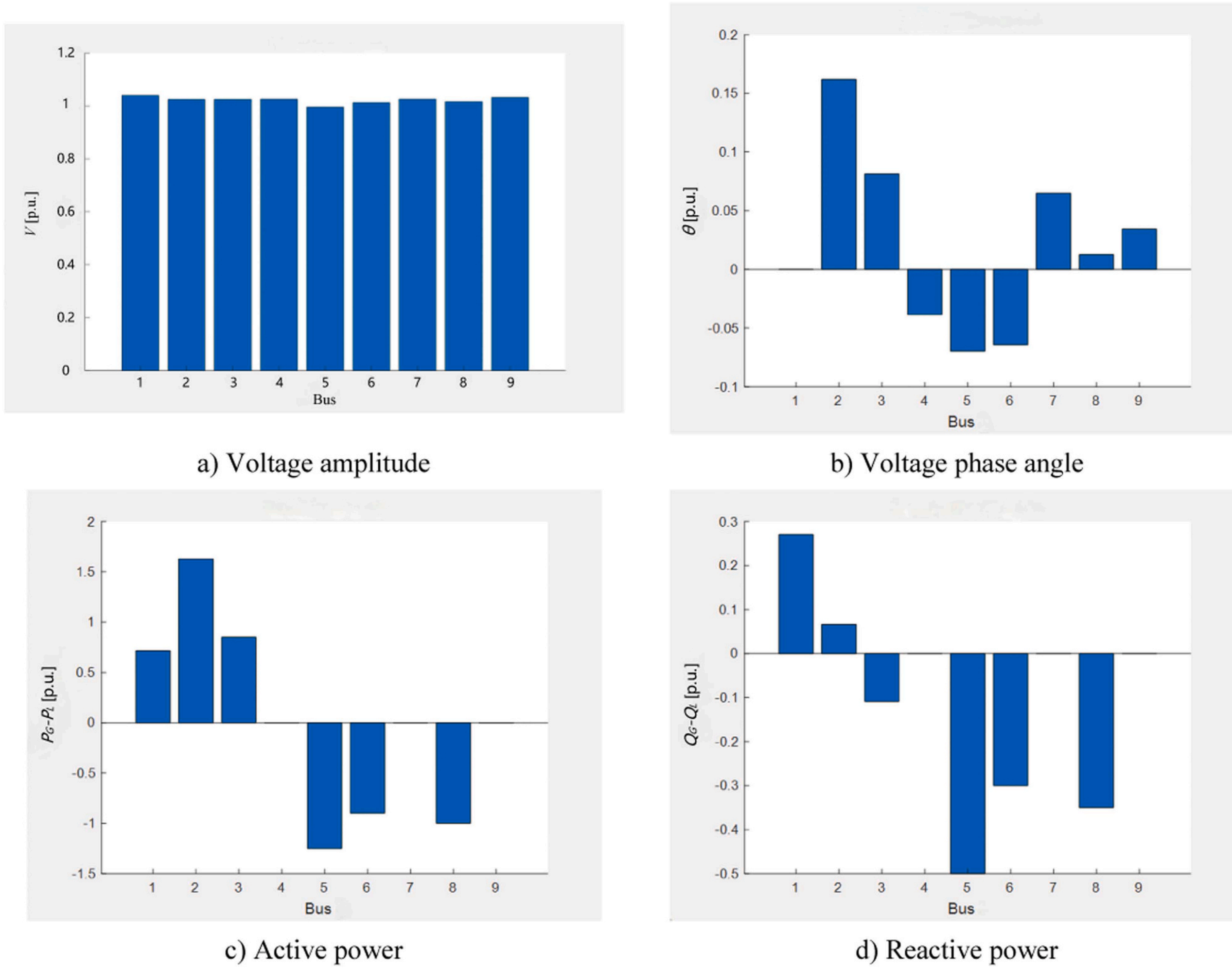

Fig. 4. Power flow analysis results.

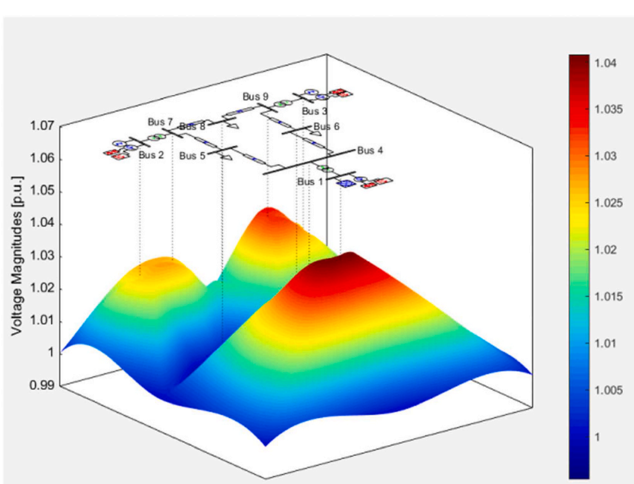

Fig. 5. Voltage distribution of test system.

calculate the injection and absorption of reactive power at each bus, facilitating an examination of the sources and destinations of reactive power for tracking its flow within the system. Based on this analysis, a load flow calculation is employed to determine the reactive power flow at each bus and quantify the level of coupling between system buses. The resulting coupling degree indices are then normalized for subsequent analysis and comparison. A regional load-based model for reserve capacity of reactive power is established, taking into consideration factors such as load variation and demand. Capacitors or inductors serving as reserve devices for reactive power are analyzed with respect to their capacity and optimal placement. By integrating this model with a partitioning scheme obtained through bus aggregation, the proposed index for each region's reactive power is calculated. Finally, considering factors such as reliability, capacity, location, and robustness analysis of support devices in each partitioned area, a comprehensive assessment of voltage stability in the power grid is conducted using both the proposed index for reactive power and robustness metrics.

4. Analysis of *L* index and energy function-based voltage stability index

4.1. L Index

The *L* index is a metric utilized for online monitoring of voltage stability in power systems, providing values ranging from 0 to 1. By leveraging information derived from normal power flows, the L index facilitates a straightforward and efficient computation of real-time voltage stability reserve. However, its drawback lies in its limited precision. As both active and reactive power buses approach their maximum loads, the system gradually approaches its stability boundary, resulting in an *L* index value of 1. Nevertheless, during actual simulations, not all bus *L* index values converge to the theoretical value of 1 at the system's stability boundary. Instead, certain load bus index values

Table 1
Proposed quantitative index with changing wind speeds.

Bus	8 m/s	9 m/s	10 m/s	11 m/s	12 m/s	13 m/s	14 m/s	15 m/s
4	-0.4732	-0.4765	-0.4797	-0.4828	-0.4877	-0.4916	-0.5000	-0.5027
6	-0.3765	-0.4231	-0.4737	-0.5890	-0.6551	-0.8087	-1.0000	-1.1143
7	-0.6979	-0.6999	-0.7019	-0.7038	-0.7075	-0.7110	-0.7142	-0.7159
8	-0.6791	-0.6904	-0.7014	-0.7119	-0.7321	-0.7512	-0.7692	-0.7779
9	-0.6015	-0.6044	-0.6072	-0.6099	-0.6152	-0.6202	-0.6250	-0.6273

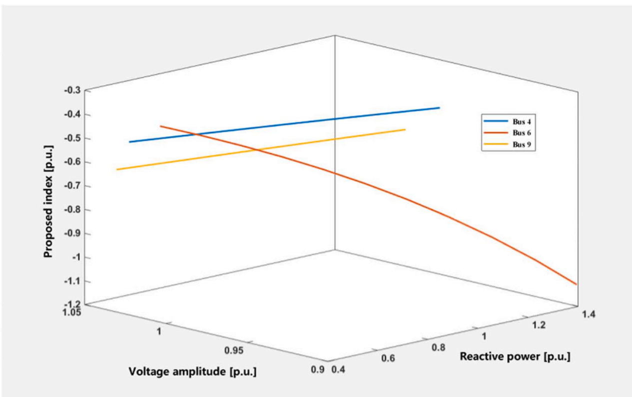

Fig. 6. Relation of proposed index, voltage amplitude and reactive power.

exceed 1 while others fall below it, indicating inherent limitations in the ability of the L index to accurately differentiate between voltage stability levels among load buses

When evaluating the voltage stability of multi-bus systems using the L index, it is essential to categorize the system buses into distinct groups: load buses are denoted by one set, while power supply buses are represented by another set. Once all buses have been appropriately grouped, the relationship between bus voltages and currents can be described as,

$$\begin{bmatrix} \boldsymbol{I_L} \\ \boldsymbol{I_G} \end{bmatrix} = \begin{bmatrix} \boldsymbol{Y_{LL}} \boldsymbol{Y_{LG}} \\ \boldsymbol{Y_{GL}} \boldsymbol{Y_{GG}} \end{bmatrix} \begin{bmatrix} \boldsymbol{V_L} \\ \boldsymbol{V_G} \end{bmatrix} \tag{15}$$

where $\boldsymbol{I_L}$ and $\boldsymbol{V_L}$ respectively represent the current and voltage vectors of load buses, $\boldsymbol{I_G}$ and $\boldsymbol{V_G}$ respectively represent the current and voltage vectors of power supply buses.

Through transformation, this can be obtained by,

$$\begin{bmatrix} \boldsymbol{V_L} \\ \boldsymbol{I_G} \end{bmatrix} = \begin{bmatrix} \boldsymbol{Z_{LL}} & -\boldsymbol{Z_{LL}} \boldsymbol{Y_{LG}} \\ \boldsymbol{Y_{GL}} \boldsymbol{Z_{LL}} & \boldsymbol{Y_{GG}} \end{bmatrix} \begin{bmatrix} \boldsymbol{I_L} \\ \boldsymbol{V_G} \end{bmatrix} \tag{16}$$

The load participation factor is expressed as,

$$\boldsymbol{F_{LG}} = -\boldsymbol{Z_{LL}} \boldsymbol{Y_{LG}} \tag{17}$$

Define the local voltage stabilization index L index as,

$$L_j = \left| 1 - \frac{\sum\limits_{k \in \alpha_G} F_{jk} \dot{V}_k}{\dot{V}_j} \right| \tag{18}$$

where L is an assessment index with a range,

$$R = \{L | 0 \leq L \leq 1\} \tag{19}$$

The linearization of data within the power system at this juncture, whether in the context of fluctuations under steady-state conditions or numerical dramatic changes during transients, clearly deviates from the actual system characteristics. Such linearization should be considered merely as an auxiliary factor.

4.2. *Voltage stability assessment index based on energy function*

Traditional voltage stability refers to the property of bus voltages returning to a stable constrained range after experiencing disturbances. Initially, this was determined by establishing a threshold value for local system voltage collapse, providing a binary conclusion of stability or instability based on the current voltage state. However, with advancements in systems analysis, the assessment of voltage stability has evolved to quantify the characteristics of the power system as it approaches the critical collapse point following specific disturbances. This characteristic emphasizes capturing the transition process from voltage stability to critical instability. Quantifying this characteristic necessitates not only determining the distance between the current and critical voltage operating points but also understanding how changes in control variables impact overall voltage security across the entire system range. From an energy perspective, we propose a method for evaluating system voltage stability that considers variations in reactive power injection within the system and their effects on both voltage stability reserve and trend, thereby enabling comprehensive assessment of voltage stability. The relationship between reactive current, voltage, and energy is depicted in Fig. 2, as revealed by the energy function model.

The power balance in an electrical system encompasses both active and reactive power balances, with the former closely associated with voltage phase angle and the latter primarily dependent on voltage magnitude. To describe this physical characteristic, a heuristic energy function is employed by integrating the equation for active power balance with respect to the phase angle and incorporating the reactive component based on voltage magnitude. The expression for this heuristic energy function can be represented as,

$$E = \int_{(\theta^0, U^0)}^{(\theta, U)} [f(\theta, U), g(\theta, U)] \begin{bmatrix} d\theta \\ dU \end{bmatrix} \tag{20}$$

$$E_i \int_{(\theta_i^0, U_i^0)}^{(\theta, U_i)} [f_i(\theta_i, U_i), g_i(\theta_i, U_i)] \begin{bmatrix} d\theta_i \\ dU_i \end{bmatrix} \tag{21}$$

where U_i represents the voltage magnitude of bus i, U_i^0 denotes its initial steady-state value, θ_i represents the phase angle of bus i, θ_i^0 denotes its

Table 2
L index with changing wind speeds.

Bus	8 m/s	9 m/s	10 m/s	11 m/s	12 m/s	13 m/s	14 m/s	15 m/s
4	0.7639	0.7618	0.7602	0.759	0.7565	0.7542	0.8771	0.7485
6	0.8118	0.7885	-0.2368	-0.694	0.6725	0.5957	-1.5217	-1.8449
7	0.6515	0.6501	0.6491	0.649	0.6463	0.6445	0.8223	0.6425
8	0.6605	0.6548	0.6500	0.6441	0.6341	0.6253	0.8122	0.6111
9	0.6993	0.6978	0.6970	0.6951	0.6932	0.3821	0.8462	0.6864

initial steady-state value, and E_i represents the energy of bus i.

The expressions for f_i and Q_i are represented as,

$$f_i\left(\theta_i, U_i\right) = P_i - \sum_{j=1}^{n} B_{ij}|U_i||U_j|\sin(\theta_i - \theta_j) - \sum_{j=1}^{n} G_{ij}|U_i||U_j|\sin(\theta_i - \theta_j) \tag{22}$$

$$g_i\left(\theta_i, U_i\right) = U_i^{-1}\left[Q_i + \sum_{j=1}^{n} B_{ij}|U_i||U_j|\cos(\theta_i - \theta_j)\right] - U_i^{-1} \times \sum_{j=1}^{n} G_{ij}|U_i||U_j|\sin(\theta_i - \theta_j) \tag{23}$$

where G_{ij} and B_{ij} represent the conductance and susceptance between buses i and j, respectively.

The active power balance equation for bus i is expressed as,

$$P_{Li} = U_i \sum_{j=1}^{n} U_j\left[G_{ij}\cos(\theta_i - \theta_j) + B_{ij}\sin(\theta_i - \theta_j)\right] \tag{24}$$

The reactive power balance equation in the absence of compensation is formulated as,

$$Q_{Li} = U_i \sum_{j=1}^{n} U_j\left[G_{ij}\sin(\theta_i - \theta_j) + B_{ij}\cos(\theta_i - \theta_j)\right] \tag{25}$$

Therefore, the expression of the energy function without considering reactive power compensation can be obtained as:

$$\begin{aligned} E_i = & \int_{(\theta_i^0, U_i^0)}^{(\theta, U_i)} [f_i, g_i] \begin{bmatrix} d\theta_i \\ dU_i \end{bmatrix} = \\ & P_{Li}(\theta_i - \theta_i^0) + \frac{1}{2}\sum_{j=1}^{n} \mathrm{B}_{ij}|\mathrm{U}_i^0|\left|\mathrm{U}_j^0\right|\cos\left(\theta_i^0 - \theta_j^0\right) - \\ & \frac{1}{2}\sum_{j=1}^{n} B_{ij}|U_i||U_j|\cos(\theta_i - \theta_j) - Q_{Li}\ln(U_i/U_i^0) - \\ & (\mathrm{U}_i^0)^{-1}\sum_{j=1}^{n} \mathrm{G}_{ij}|\mathrm{U}_i^0|\left|\mathrm{U}_j^0\right| sin(\theta_i^0 - \theta_j)(\mathrm{U}_i - \mathrm{U}_i^0) - \\ & \sum_{j=1}^{n} \mathrm{G}_{ij}|\mathrm{U}_i^0|\left|\mathrm{U}_j^0\right| cos\left(\theta_i^0 - \theta_j^0\right)(\theta_i - \theta_i^0) \end{aligned} \tag{26}$$

The influence of the reactive power injection set Q on the energy E is double, on the one hand, through the influence on the bus voltage U to indirectly affect the energy E, in the middle of the current calculation to realize the connection between Q and U. On the other hand, directly as a variable to affect the energy E, and directly through the energy function model to connect.

Thereby, the energy function can be expressed uniformly as a relation:

$$E = E(Q, U, \theta) \tag{27}$$

Meanwhile, to quantify the extent of voltage stabilization, we define the effective energy ΔE as the energy of the bus influenced by reactive power variation that ensures system voltage remains within the stabilization limit. The voltage at the stabilization level is denoted as U_0, while U_α represents the critical value at which voltage decreases. Drawing upon the physical significance of Q-ln$U$ curve, we express ΔE as,

$$\Delta E = E_0(Q_0, U_0, \theta_0) - E_{\mathrm{cr}}(Q_{\mathrm{cr}}, U_{\mathrm{cr}}, \theta_{\mathrm{cr}}) \tag{28}$$

According to the aforementioned equation, taking into account the quantitative requirements for voltage stability, the voltage stability proposed index of reactive power injection can be defined as,

$$D = E/\Delta E \tag{29}$$

Specifically for bus i, when the stability reserve D is between −1 and 0, the voltage is stable. When $D = -1$, the voltage is at the destabilization threshold. When D is less than −1, the voltage is destabilized. And when D is greater than 1, the voltage rises further based on the initial value of the steady state.

The index is modelled by the energy function model. It can dynamically adjust the reactive power injection and consumption to adapt to various wind output situations. Thus, the method provides accurate voltage stability assessment, with the variability and intermittency of wind energy sources. Moreover, the validity and reliability of the proposed method is verified in case studies under different wind speed conditions, of which the results show the robustness and adaptability of the proposed index. Furthermore, the energy function-based method not only assesses current voltage stability, but also predicts the impact of future wind changes on the system. Therefore, it can advise grid operators on preventive measures to minimize the negative impacts of wind fluctuations.

The L index relies on linearization, which produces large errors when the system tends to a critical state. Thus, as for the predictive accuracy, the L index is prone to errors when the system is close to a critical point and has a relatively low accuracy. In contrast, the proposed energy function-based method has better performances in taking into account the system nonlinear characteristics and dynamic changes. With high loads and fluctuating wind power, the proposed scheme is able to assess the voltage stability more accurately.

As for the computational efficiency, the proposed energy function-based method relies on the power flow calculation. The computational time of the proposed scheme is mainly determined by the power flow calculation, which is acceptable for online application. There is no doubt that the L index has higher computational efficiency because of its linearization.

When it comes to the real-time grid management, although the L index has higher computational efficiency, its relatively low accuracy hinders its application. Comparatively, the proposed energy function-based approach has higher accuracy and acceptable computational efficiency. Therefore, the proposed approach can be used in real-time grid management.

The assessment results are useful for the management of reactive power, and the long-term energy planning and reliability assessments. Through energy function and power flow analysis, the proposed assessment approach can identify key buses that require reactive power compensation. Thus, the management of reactive power can be activated and compensate reactive power at these key buses. This helps to ensure grid voltage stability and reduce the impact of voltage fluctuations caused by the change of load and wind power.

Moreover, as for the reliability assessments, the proposed approach can select the buses where the reactive power is required more frequently. These buses can be regarded as the vulnerable part of the power system, and have low reliability. And the reactive power compensation device is suggested to be deployed in these buses, when it comes to long-term energy planning.

5. Case studies

5.1. Test system description

The IEEE-9 bus system is used for analysis, where a wind farm is deployed at Bus 6, as shown in Fig. 3. The voltage, reactive power and active power of each bus in the system are collected for voltage stability assessment. The capacity of the wind farm is 200 MW and the power factor is set as 0.99. The power flow analysis results and voltage distribution of the test can be demonstrated in Figs. 4 and 5, respectively.

5.2. Impact of wind power variation

The wind speeds vary from 8 m/s to 15 m/s, and the voltage stability index of the proposed approach are calculated, as shown in Table 1. The wind speed is the critical factor that influenced the stability assessments. As the wind speed increases 8 m/s to 15 m/s, the output active power and reactive power of the wind farm changes, which influences the voltage stability margin of the test system. It is clear that the proposed approach is able to reflect the change of the voltage stability margin as the wind speed increases.

The system voltage stability index decreases to −1 as the wind power output gradually increases, indicating a reduction in the voltage stability margin at each bus of the system. Notably, there is a greater variation in the index observed at bus 6, suggesting that with an increase in wind farm output, the voltage at this particular bus experiences a more rapid decline, exacerbating overall voltage stability. Bus 6 serves as the integration point for wind energy and can be identified as a weak bus within the system. This vulnerability can be attributed to the wind farm providing reactive power support to the connection point when operating at lower wind power outputs. However, this support diminishes as wind power output increases due to increased reactive power consumption by nearby buses and access points. Consequently, it becomes challenging for the wind turbine to maintain sufficient reactive power support, leading to decreased voltage stability at bus 6. Fig. 6 illustrates graphical results depicting changes in voltage stability index for bus 6 and neighboring buses 4 and 9.

As the output of the wind farm at bus 6 gradually increases, there is a corresponding decrease in voltages at buses 6, 4, and 9 along with an increase in reactive power consumption. Consequently, the voltage stability proposed index based on reactive power injection also gradually decreases. When the reactive load exceeds a certain threshold, the proposed index at bus 6 drops below −1, indicating voltage instability and necessitating additional reactive compensation. This validates the efficacy of utilizing energy-based voltage stability proposed index for analyzing changes in stability reserves at vulnerable buses during wind power integration scenarios.

5.3. Comparison to L index

With increasing wind speeds, the *L* index can be calculated, as shown in Table 2. However, the *L* index fails to reflect the change of the voltage stability margin and the assessment results varies significantly when the system tends to a critical state, because it relies on linearization.

By analyzing the comparative data between the proposed index and *L* index, and observing their behavior under identical wind speed conditions at the same bus, several conclusions can be drawn. Firstly, both indices partially reflect the increasing fragility of overall system stability as wind power penetration increases. This aligns with the previously mentioned decrease in voltage stability as wind power output gradually rises. Additionally, the L index also indicates changes in special bus 6 selected earlier, suggesting a faster deterioration of voltage stability. Bus 6 can be identified as a weak bus due to its connection with wind power, as explained in detail before.

However, it is worth noting that compared to the *L* index, the proposed index shows relative improvement. Specifically, under extreme system conditions, there is a significant deviation observed in the *L* index. This discrepancy can be attributed to limitations associated with linearization treatment used for calculating the *L* index itself. Thus, it should only serve as an initial reference point. However, during limit stable states, nonlinear characteristics of this particular bus surpass assumptions made during linearization treatment process leading to increased errors

6. Conclusion

As power systems become increasingly intricate, there is an urgent imperative for comprehensive research on the impact of reactive power on static voltage stability. The analysis delves into the influence of reactive power on voltage stability, employing a combination of energy functions and power flow analysis to propose a quantitative index for assessing static voltage stability. Case studies were conducted to demonstrate the efficacy of this proposed approach. The proposed assessment approach can identify key buses that require reactive power compensation. The management of reactive power can be activated and compensate reactive power at these key buses. This helps to ensure grid voltage stability and reduce the impact of voltage fluctuations caused by the change of load and wind power. Therefore, considering the varies of the renewable energies, it is recommended to fast identify the key buses of voltage stability, and compensate reactive power at these key buses. In addition, the proposed approach can select the buses where the reactive power is required more frequently. These buses can be regarded as the vulnerable part of the power system, and have low reliability. And the infrastructure of reactive power compensation is suggested to be deployed in these buses.

The proposed energy function-based method relies on power flow calculation, which needs the parameters of the power system. However, as the penetration of renewable energies increase, the detailed parameters of the power system is becoming inaccessible gradually. Therefore, in future studies, the data-driven voltage stability assessment approach will be presented.

CRediT authorship contribution statement

Yongji Cao: Supervision, Conceptualization. **Ruicong Ma:** Software. **Jian Zhang:** Writing – original draft. **Changgang Li:** Supervision. **Shihao Zou:** Visualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper

Data availability

Data will be made available on request.

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